Credit Card Loan Risk Analysis

Exploratory data analysis of Geldium customers

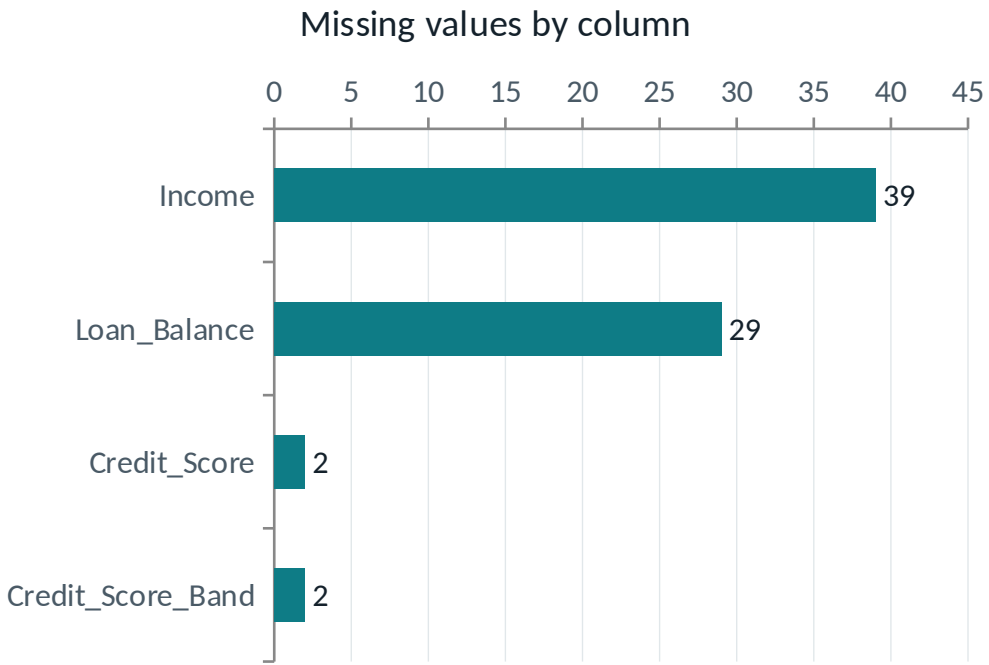


Objective and dataset

500	23	16%	4.2%
Customers	Columns	Delinquent	High risk flag

- Goal is to find customers at higher risk of missing payments
- Source file is Geldium_PowerBI_Data.csv
- Columns cover profile, credit behaviour, loan details and six months of payment status

Data quality and cleaning



What we did

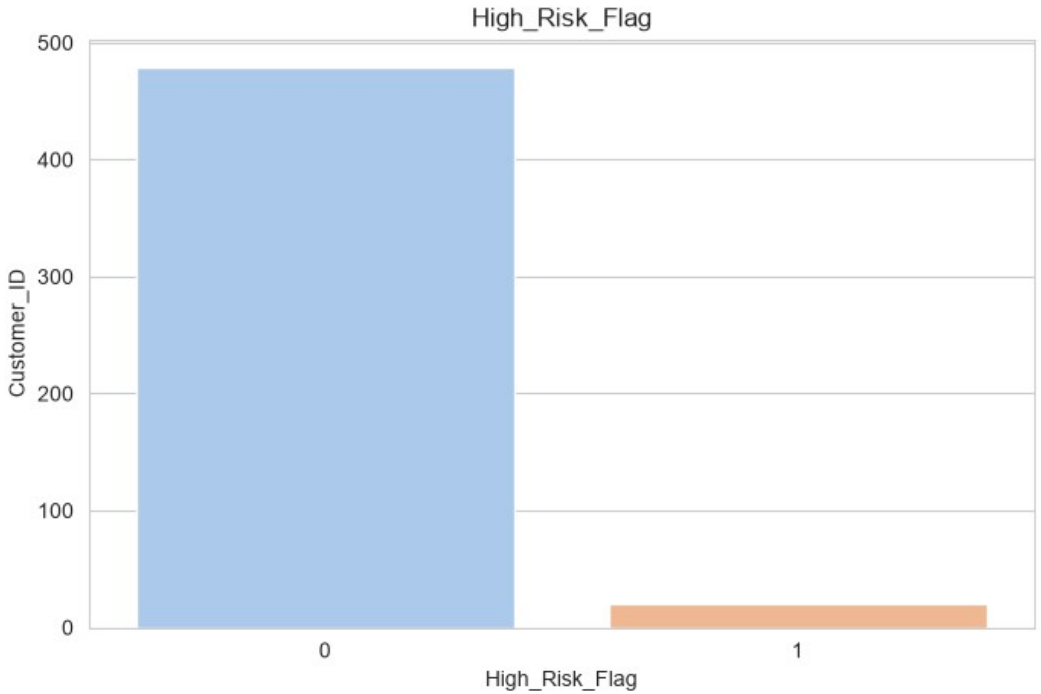
- No duplicate rows found
- Median used to fill Income, Loan_Balance and Credit_Score
- Credit_Score_Band needs the most common value instead
- Utilization capped at 1

Summary statistics

Feature	Mean	Median	Min	Max
Age	46.3	46.5	18	74
Income	108,380	107,658	15,404	199,943
Credit Score	577.7	586	301	847
Credit Utilization	0.49	0.49	0.05	1.03
Missed Payments	2.97	3	0	6
Loan Balance	48,654	45,776	612	99,620
Debt to Income Ratio	0.30	0.30	0.10	0.55
Account Tenure (years)	9.7	10	0	19

Average credit score of 578 sits in the poor range, so many customers already carry credit risk

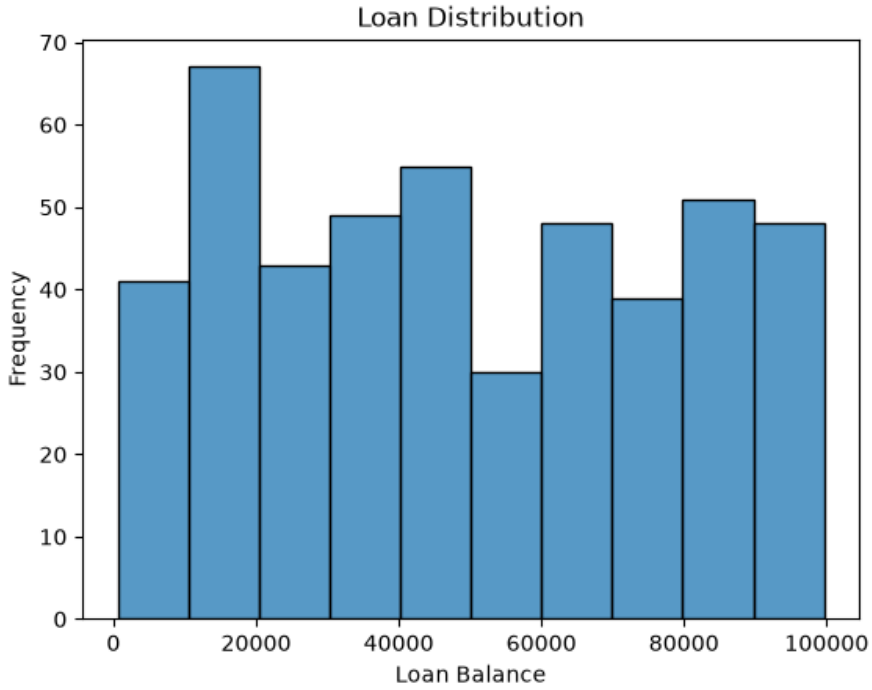
High risk customers



What we see

- Only about 4.2% carry the high risk flag, around 21 customers
- Data is very imbalanced, so a model needs class weights or resampling

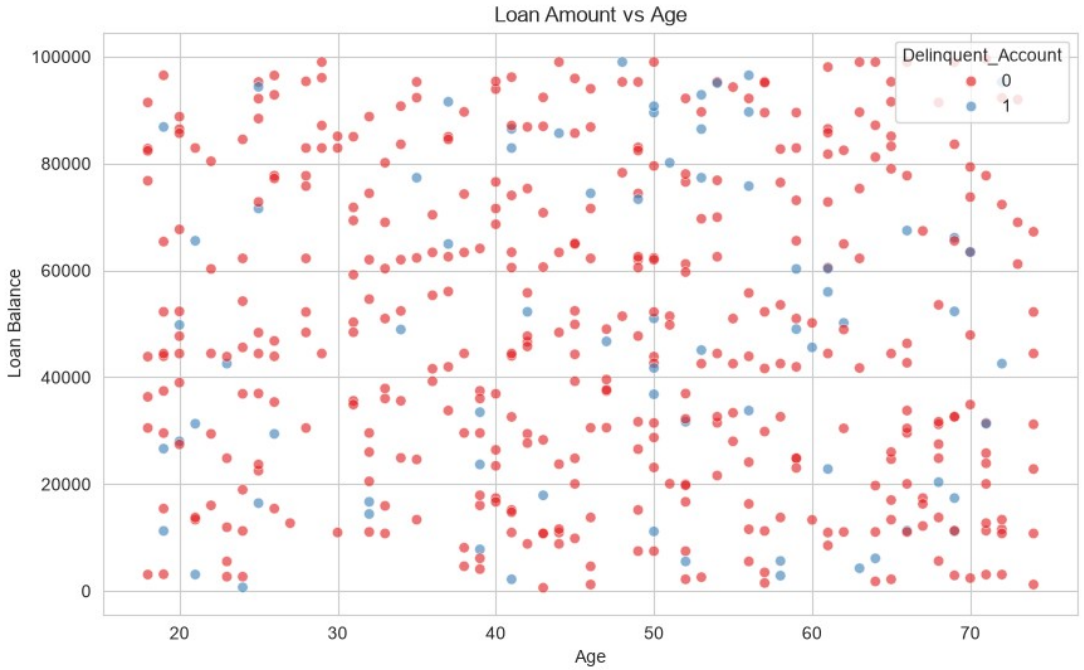
Loan balance spread



What we see

- Balances range from near 0 to 100,000
- Most common band is 10,000 to 20,000, then 40,000 to 50,000

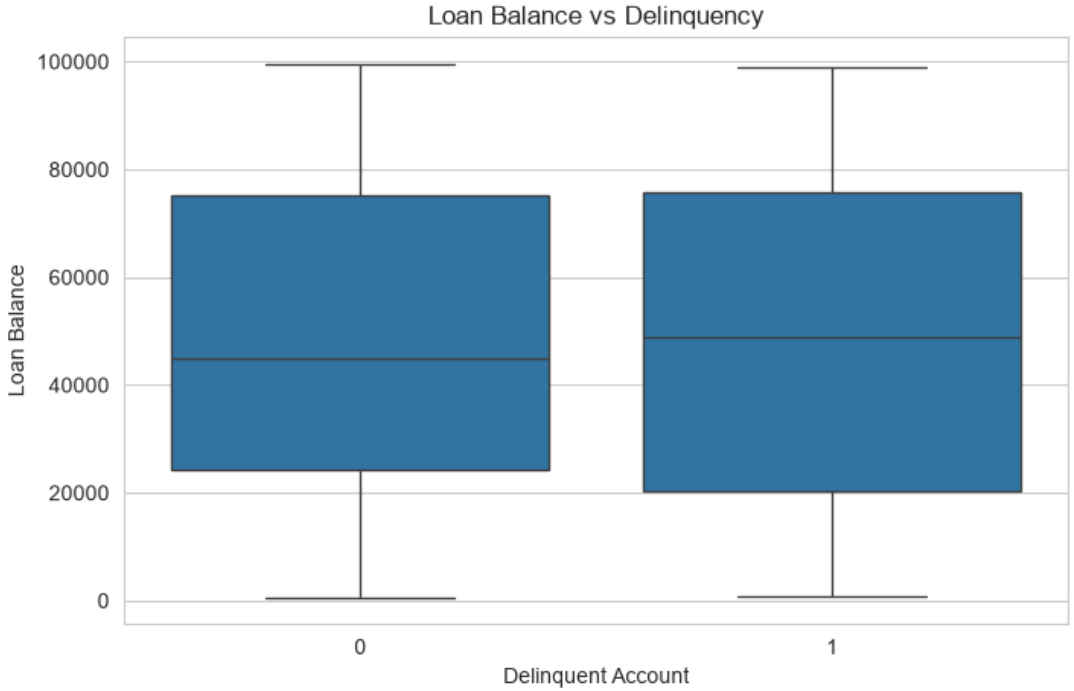
Loan balance and age



What we see

- Delinquent accounts appear at every age and loan size
- Age and balance alone do not separate risky customers

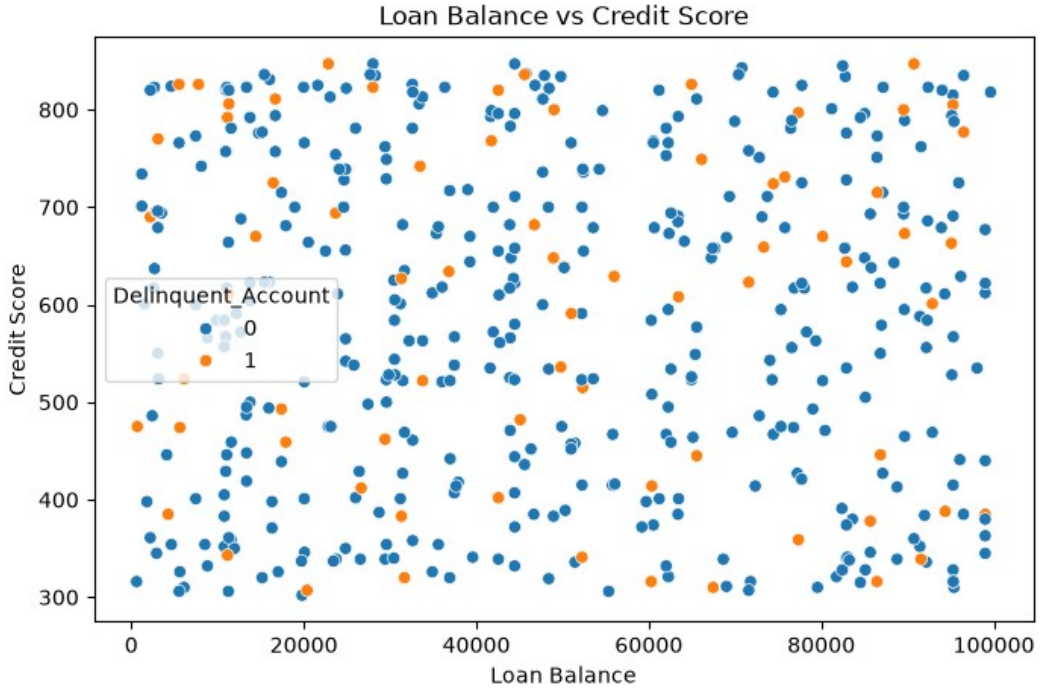
Balance and delinquency



What we see

- Delinquent accounts have a slightly higher median, near 49,000 against 45,000
- Boxes overlap a lot, so balance is a weak signal alone

Balance and credit score



What we see

- Delinquent accounts are found in every credit score range
- No clean cut line, so combine several signals

Key findings

1. About 16% of customers are delinquent, a strong reason for early monitoring
2. No single factor drives delinquency, each one is weak alone
3. Key indicators are missed payments, credit score, utilization, balance, income and debt to income
4. Missed payments average almost 3 per customer
5. Only 4.2% are flagged high risk, so the flag may miss many delinquent customers

Risk based collection plan

High risk

Who
Several warning signs together, such as many missed payments, low credit score and high utilization

Action
Early calls and personal follow up

Medium risk

Who
One or two warning signs

Action
Reminders and regular follow up

Low risk

Who
Clean payment history

Action
Standard collection process

Conclusion and next steps

Delinquency comes from a mix of factors, so a risk based plan will find risky customers earlier and use collection effort better

- Check correlation of missed payments, utilization and debt to income with delinquency
- Compare delinquency by employment, card type, location and score band
- Build a logistic regression or random forest model
- Share results in a Power BI dashboard